ARYAN SINGH

# MINOR PROJECT

## PROBLEM STATEMENT

**Breast Cancer Prediction Using Support Vector Machine (SVM)**

Breast cancer is one of the most common and life threatening disease found in women. The symptoms can be recognized at early stages and to correctly detect whether a person is diagnosed with breast cancer or not i will be using supervised learning algorithm.

**Project Objective**

The main objectives of this project are:

- To develop a supervised machine learning model for breast cancer classification.

- To preprocess the dataset by handling missing values (if any), encoding labels, and scaling numerical features.
- To train a **Support Vector Machine (SVM)** classifier using the prepared dataset.
- To evaluate the model using classification metrics such as Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.
- To demonstrate how machine learning can support early detection of breast cancer and assist healthcare professionals in making informed decisions.

DATASET:

https://www.kaggle.com/datasets/uciml/breast-cancer-wisconsin-data

For this project, the **Breast Cancer Wisconsin (Diagnostic) Dataset** was selected. This is a widely used benchmark dataset for binary classification problems in machine learning. It contains diagnostic measurements obtained from digitized images of **Fine Needle Aspiration (FNA)** tests performed on breast masses. The objective is to classify each tumor as either **Benign (B)** or **Malignant (M)** based on various cellular characteristics.

The dataset contains **569 patient records**, each described by **30 numerical features** computed from cell nuclei measurements. The target variable represents the diagnosis:

- **M** – Malignant (Cancerous)
- **B** – Benign (Non-cancerous)

### 3.1 Handling Missing Values

Missing values can reduce the accuracy of a machine learning model if they are not handled properly. Therefore, the dataset was examined to identify any null or missing entries.

After inspection, it was found that the **Breast Cancer Wisconsin (Diagnostic) Dataset contains no missing values**. All **569 records** had complete information for each of the **30 numerical features**. Hence, no techniques such as mean, median, or mode imputation were required.

**Outcome:**

- Missing values found: **0**
- Action Taken: **No preprocessing required for missing values.**

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### 3.2 Removing Duplicates

Duplicate records can introduce bias into the model by giving repeated observations more importance than others. Therefore, the dataset was checked for duplicate rows.

The dataset was examined for duplicate entries, and any duplicate records, if present, were removed before model training. This ensured that every observation represented a unique patient.

**Outcome:**

- Duplicate records were checked and removed to maintain data quality.

- Final dataset contained only unique observations.

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### 3.3 Feature Encoding

Machine learning algorithms require numerical input. The dataset contains one categorical feature, **Diagnosis**, which is the target variable.

The diagnosis labels were converted into numerical values using **Label Encoding**:

| Original Value | Encoded Value |
|---|---|
| Benign (B) | 0 |
| Malignant (M) | 1 |

All other features, such as radius, texture, perimeter, area, smoothness, compactness, concavity, symmetry, and fractal dimension, were already numerical and therefore required no additional encoding.

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### 3.4 Feature Scaling (Normalization)

The Breast Cancer dataset contains features measured on different scales. For example, **Area Mean** has much larger values than **Smoothness Mean** or **Fractal Dimension Mean**.

Since the **Support Vector Machine (SVM)** algorithm is sensitive to feature magnitude, feature scaling was necessary.

**Standardization (StandardScaler)** was applied to transform each feature so that:

- Mean = **0**
- Standard Deviation = **1**

This preprocessing step ensures that every feature contributes equally to the model and improves the performance and convergence of the SVM classifier.

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## 3.5 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the dataset before training the machine learning model.

#### Dataset Overview

- **Dataset Name:** Breast Cancer Wisconsin (Diagnostic)
- **Total Records:** 569
- **Total Features:** 30 numerical features
- **Target Variable:** Diagnosis
- **Classification Type:** Binary Classification

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#### Class Distribution

The target variable consists of two classes:

| Diagnosis | Meaning | Number of Samples |
| --- | --- | --- |

| | | |
|---|---|---|
| Benign (B) | Non-Cancerous | 357 |
| Malignant (M) | Cancerous | 212 |

The dataset contains more benign cases than malignant cases, but the class distribution is still reasonably balanced for training a classification model.

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#### Feature Types

The dataset contains only **numerical features**, which are grouped into three categories:

- Mean values (e.g., Radius Mean, Texture Mean)
- Standard Error values
- Worst (largest) values

## 6. Model Used

## Support Vector Machine (SVM)

The **Support Vector Machine (SVM)** is a supervised machine learning algorithm used for both classification and regression tasks. In this project, SVM was used as a **binary classification** algorithm to predict whether a breast tumor is **Benign (B)** or **Malignant (M)** based on diagnostic features.

SVM works by finding an optimal decision boundary, known as a **hyperplane**, that separates data points belonging to different classes. The algorithm selects the hyperplane that maximizes the margin between the two classes, which helps improve the model's ability to classify unseen data accurately.

Since the Breast Cancer Wisconsin dataset contains numerical features and is suitable for binary classification, SVM is an effective choice due to its high accuracy and strong generalization capability.

### Advantages of SVM

- Effective for binary classification problems.
- Performs well on datasets with many numerical features.
- Maximizes the margin between classes, improving generalization.
- Less prone to overfitting when properly configured.
- Produces high classification accuracy on medical datasets.

**Disadvantages of SVM**

- Sensitive to the scale of input features, making feature scaling necessary.
- Training time increases for very large datasets.
- Selecting appropriate parameters (kernel, C, gamma) may require experimentation.

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# 7. Training Process

The following steps were followed to train the Support Vector Machine model:

**Step 1: Data Collection**

The Breast Cancer Wisconsin (Diagnostic) dataset containing **569 patient records** was collected from the UCI Machine Learning Repository.

**Step 2: Data Preprocessing**

Before training, the dataset was preprocessed by:

- Checking for missing values.
- Removing duplicate records.
- Encoding the diagnosis labels (Benign = 0, Malignant = 1).
- Standardizing all numerical features using StandardScaler.

**Step 3: Splitting the Dataset**

The dataset was divided into two parts:

- **Training Set:** 80%
- **Testing Set:** 20%

The training dataset was used to learn patterns from the data, while the testing dataset was used to evaluate the model's performance on unseen samples.

**Step 4: Model Training**

The Support Vector Machine classifier was trained using the training dataset. During training, the algorithm identified the optimal hyperplane that best separated benign and malignant tumor samples based on their feature values.

**Step 5: Model Prediction**

After training, the model predicted the diagnosis of tumor samples in the testing dataset. These predictions were then compared with the actual diagnoses to evaluate the model's performance.

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## 8. Evaluation Metrics and Results

The performance of the Support Vector Machine model was evaluated using standard classification metrics.

### Accuracy

Accuracy measures the proportion of correctly classified samples among all predictions.

**Formula:**

**Accuracy = (TP + TN) / (TP + TN + FP + FN)**

**Result: 97.37%**

The high accuracy indicates that the model correctly classified most tumor samples.

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### Precision

Precision measures how many of the samples predicted as malignant were actually malignant.

**Formula:**

**Precision = TP / (TP + FP)**

**Result: 97.56%**

A high precision value indicates that the model made very few false-positive predictions.

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### Recall

Recall measures how many actual malignant tumors were correctly identified.

**Formula:**

**Recall = TP / (TP + FN)**

**Result: 95.24%**

A high recall is important in medical diagnosis because it minimizes the number of missed cancer cases.

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#### F1-Score

The F1-score is the harmonic mean of precision and recall.

**Formula:**

**F1 = 2 × (Precision × Recall) / (Precision + Recall)**

**Result: 96.39%**

The F1-score indicates that the model achieves a good balance between precision and recall.

**Confusion Matrix**

The confusion matrix summarizes the prediction results.

| Actual / Predicted | Benign | Malignant |
|---|---|---|
| **Benign** | 71 | 1 |
| **Malignant** | 2 | 40 |

Interpretation:

- **71** benign tumors were correctly classified.
- **40** malignant tumors were correctly classified.
- **1** benign tumor was incorrectly classified as malignant.
- **2** malignant tumors were incorrectly classified as benign.

The confusion matrix shows that the model performs exceptionally well in distinguishing between benign and malignant tumors.

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**Overall Performance**

The Support Vector Machine classifier achieved excellent performance on the Breast Cancer Wisconsin dataset.

| Evaluation Metric | Value |
|---|---|
| Accuracy | **97.37%** |
| Precision | **97.56%** |
| Recall | **95.24%** |
| F1-Score | **96.39%** |

These results demonstrate that the model is highly effective for breast cancer classification.

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## 9. Conclusion

This project successfully developed a **Support Vector Machine (SVM)** model to classify breast tumors as **benign** or **malignant** using the Breast Cancer Wisconsin (Diagnostic) dataset. The dataset was preprocessed by checking for missing values, removing duplicate records, encoding the target labels, and standardizing numerical features. Exploratory Data Analysis provided insights into the data and confirmed its suitability for supervised learning.

After training and evaluation, the SVM classifier achieved an **accuracy of approximately 97.37%**, along with high precision, recall, and F1-score. The confusion matrix showed that only a few samples were misclassified, indicating that the model can reliably distinguish between benign and malignant tumors.

Overall, the project demonstrates that supervised machine learning, particularly the Support Vector Machine algorithm, can effectively support the early detection of breast cancer and assist healthcare professionals in making accurate diagnostic decisions.

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## 10. References

1. *Breast Cancer Wisconsin (Diagnostic) Data*. Available at: https://www.kaggle.com/datasets/uciml/breast-